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Behavioural analysis of independent value-based learning in non-cooperative games

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ABSTRACT

Multi-agent reinforcement learning has received increased attention in cooperative games. However, research in non-cooperative games is lagging behind. Independent value-based learning algorithms have demonstrated simplicity and versatility in various contexts. In this paper, we study the behavior of these algorithms in non-cooperative settings. We explain the conditions that a game must satisfy for the algorithms to work. We further test the algorithms in our proposed game Food Chain that simulates an ecosystem. Our results show that independent value-based learning algorithms can converge to Nash equilibrium, only when the Nash equilibrium consists of uniformly random policies over the feasible actions.

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Multi-agent reinforcement learning; independent value-based learning; game theory; non-cooperative games; Nash equilibrium

1. Introduction

Recent multi-agent reinforcement learning (MARL) research has mostly focused on fully cooperative games [1–3] where there is no conflict between agents as they share a common reward. When agents are allowed to share information, we can apply single-agent reinforcement learning (RL) algorithms to a fully cooperative game where the collective agent tries to find the optimal joint action, given the joint observation, although the problem is still challenging in practice because of what the computational complexity of the joint action brings. In a traditional single-agent setting, the agent needs to learn which action gives the best reward. A pure strategy that maximizes expected reward is guaranteed to exist in fully cooperative finite games, meaning that the agents can always deterministically choose the optimal action that has the highest payoff.

Non-cooperative environments, however, involve conflicts that make learning more complicated. Agents may belong to different parties that would like to have private control to protect their own interests. Rational and self-interested agents will have to fight for what is best for them, as increasing one agent's reward might lead to the decline of other's gains. We can no longer intuitively define the solution for a game as the optimal strategy resulting in the highest reward. One of the possible solution concepts is the Nash equilibrium

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(NE). NEs describe the stable points in games, which are associated with the convergence to optimal behavior of agents.

Many real-world problems can be modeled as non-cooperative games, such as business competition, political campaigns, stock trading, bidding in auctions, cyber security, and traffic routing. For example, if we omit the long-term economic growth, the stock market is often considered a 0-sum fully competitive game where one person's profit is another person's loss. Another example is cyber security, which must study the conflicts between attackers and defenders. These problems feature the competitive nature of games because they share one thing in common, parties with conflicting interests. Substantial research has been done on MARL in fully cooperative games, and non-cooperative games have received less attention [3, 4].

Some studies have explored the application of MARL to solve games. One of which is independent value-based learning (IVBL). IVBL is a family of algorithms that models the multi-agent environments as single-agent environments, and they compute for each agent the expected reward of each action under each state. IVBL is simple and highly scalable because it is decentralized. It has been adopted for both cooperative [5–8] and non-cooperative tasks [9–12]. This paper analyses the performance of IVBL in non-cooperative finite games. The analysis considers characteristics of the action space and the NEs of the game, combining both formal analysis and experimental evaluation in selected two-player non-cooperative games. The results show NEs with uniformly random policies is a necessary condition for stability convergence of IVBL algorithms in non-cooperative environments. These condition highly restricts the wider applicability of IVBL. Although the researchers found that IVBL can be smoothly implemented for non-cooperative games [9–12], our study showed that the performance of these algorithm can be misleading in a lot of cases. Researchers should be aware of the weaknesses that can lead to potential failure.

The main contributions of this work include the following points:

- Behavioural analysis identifying necessary conditions for a stable convergence of IVBL which can drive the choice of algorithms in non-cooperative finite games.
- Introducing a non-cooperative game that has the characteristics which challenge the agents from a game theory perspective aligned with our analysis.
- Experimental evaluation of IVBL algorithms in selected two-player scenarios in the game proposed above.

The remainder of the paper is organized as follows: The next section provides the literature review on the key features of non-cooperative games and the most popular MARL algorithms that is related to our work. We then state the formal definitions of games and other concepts discussed in this paper in Section 3. Section 4 analyses the stability of IVBL in non-cooperative games. To verify our analysis with experimental results, a new game called Food Chain is introduced in Section 5. The game will have the key features discussed in our analysis that bring specific challenges to MARL agents. Then, we present the design as well as the results of our experiments and relevant discussions in Section 6. Finally, the drawn conclusions pulled in the last section.

2. Background and literature review

This literature review focuses on key features of non-cooperative games, as well as the MARL approaches to solve these games.

2.1. Non-Cooperative games

In single-agent or fully cooperative environments, there is one reward function for all agents. Thus, the agents only need to learn which (joint) action has the highest payoff. In non-cooperative games, there may be no optimal joint action that satisfy all agents. Several problem solving concepts/approaches for non-cooperative games are discussed in game theory [13]. These include Pareto optimum, Nash equilibrium, and correlated equilibrium. Nash Equilibrium (NE), proposed by John Nash, is a renown approach for non-cooperative games [14]. NE offers a framework to characterize and analyse stability in games and describes a stability condition with which no player can benefit from deviating from their current strategy independently. The stability is associated to the optimal outcome in non-cooperative games and hence the objective agents aim for. Generally, there may only be a mixed NE, meaning that players must adopt randomized action-selection policy consisting of probability distributions over the action space. This is the mixed policy problem.

Many fully competitive games are studied in the MARL community such as Pommerman [15], Go [16], StarCraft II [17], and Dota 2 [18]. There are also some general non-cooperative environments such as hide-and-seek [19], Neural MMO [20]. PettingZoo and Melting Pot [21, 22] are two integrated MARL environments including many cooperative and non-cooperative games. Kopazs et al. [12] studied the non-cooperative game predator-and-prey in the PettingZoo library, which is closely related to the game defined in this work. However, we lack a standard benchmark for MARL algorithms in non-cooperative games due to the difficulty of measuring agents' performance in general non-cooperative games. In cooperative games, the reward simply increases as the agents do better at the game. However, this is not the case in non-cooperative games where increasing one agent's outcome often results in the drop of another agent's return. FightLadder [4] used the Elo rating system to test MARL algorithms but they were restricted to two-player zero-sum fighting games.

2.2. MARL algorithms

RL algorithms can be categorized in to model-based and model-free RL where the model refers to the Markov decision process defined by the game. Model-based algorithms access the state transition probability function and the reward function so the agents are able to predict where the actions lead to. The model can be learnt or predefined. The idea is that agents can make plans using the model instead of real interactions so the algorithm is more sample-efficient, but the cost is higher complexity of the implementation. Early studies has shown predefined model can produce highly efficient action sampling in complex games such as Go [16, 23]. More recent research demonstrated great success achieved by a learnt model [24]. Model-free agents directly learn the policy or the expected reward of state-action pairs through trial and error. Since we don't need to worry about implementing or learning the model, model-free algorithms are simple to implement but usually less

data-efficient. The algorithms discussed in this paper such as Q-learning are all model-free because the weakness of the algorithms we are going to address is particularly rooted in the learning of the expected reward of the state-action pairs that happens in model-free RL.

Independent learning (IL) algorithms that are carried over from the traditional single-agent RL domain treat other agents as part of the environment. The agents are essentially unaware of the existence of one another. IL algorithms are popular for handling many complex problems [2, 25–27]. Among these, independent value-based learning (IVBL) algorithms, such as the classic independent Q-learning (IQL) [28], learn the expected long-term cumulative return (the Q-value) of each action at each state through sampling. IVBL algorithms including IQL are often considered as baseline and natural choice in a wide range of MARL studies because of their simplicity and scalability. For example, they were studied in various cooperative environments [5–8]. The problem is that IVBL agents only help produce the average reward of each action for each agent, but do not pay attention on what the action-selection policy. This is not a problem in single-agent or cooperative environments as we can simply pick the action with the highest return. Some studies addressed this issue in non-cooperative games by simply choosing the action with the highest empirical return [9, 10, 12] and observed reasonably good outcome. Kopacz et al. [12] has examined IVBL in the non-cooperative game predator-and-prey. Both the algorithm and the game are closely related to our work. Other researchers [11] tried to address this problem using the softmax function given in Equation (1) to calculate a policy based on the payoff. Softmax takes a vector as input and computes a probability distribution over the original vector entries, such that the higher the original value, the higher the probability.

$$\text{softmax}([x_1, \dots, x_n]) = \left[\frac{e^{x_1}}{\sum_i e^{x_i}}, \dots, \frac{e^{x_n}}{\sum_i e^{x_i}} \right] \quad (1)$$

The latter approach emphasis the action with higher empirical return. However, these approaches have a problem that guarantees failure of learning in a wide range of games. We will take a closer look in the analysis section.

Unlike IL, joint action learning (JAL) algorithms allow the agent to gain access to the information processed by all other agents, such as their choices of actions, so agents can learn the expected return of joint actions. Minimax Q-learning [29] is a JAL algorithm that can be applied to two-player zero-sum games. It computes local minimax solutions for each state based on the learned joint Q-values. Nash Q-learning [30] is a similar algorithm that computes the local NE instead. It converges to a NE under highly strict assumptions that reduce the applicability of the approach to a limited set of games. Correlated Q-learning [31] computes a correlated equilibrium and has no known convergence guarantees. Zinkevich et al. [32] showed that JAL cannot converge in certain types of ill-conditioned games. Centralized learning requires full knowledge of the action spaces of all agents, which may not always be available if we are dealing with real-world problems. Another major disadvantage is that the number of joint actions is the number of possible combinations of the individual actions. This introduces combinatorics to the computational complexity of our algorithms, making them unscalable.

In contrast to the value-based approach, policy-based algorithms, such as the policy gradient methods, work around this issue by learning the policies directly, without putting an estimation on the individual action return. Policy gradient methods optimize

a parametrized policy function using gradient decent to maximize the return. An advantage of policy gradient methods is that they naturally work on continuous action spaces. The actor-critic algorithms are a family of RL algorithm which trains a parametrized policy called the actor and a state-value function called the critic to estimate the policy gradient. Soft actor-critic (SAC) [33] has become one of the most popular algorithms recently. Its multi-agent variant (MASAC) has been considered the state-of-the-art (SOTA) baseline [1, 34]. There is also naturally an independent version (ISAC) for MARL. The proximal policy optimization algorithm (PPO) [35] was based on actor-critic as well and was extended to multi-agent environments by some researchers [1, 36] (MAPPO). Surprisingly, the IL version of PPO (IPPO) has proven to substantially outperform MAPPO and other algorithms in some cooperative tasks in StarCraft II [26]. MADDPG [37] is another policy gradient algorithm that is often considered SOTA. Because it only works for continuous action spaces, we do not discuss them further in the context of finite games.

Other notable MARL algorithms that have shown good performance include VDN [38] and its improved version QMIX [39]. The weakness is that they can only work for fully cooperative games where agents share a common reward.

2.3. Analysis and discussion

As shown in this review, IVBL has been a popular selection for MARL researchers due to its simple implementation and effectiveness in various environments. However, the issue of deriving the policy from the Q-values learned by IVBL agents in non-cooperative games has not been fully addressed by the MARL community. The most intuitive solution is simply selecting the action with the highest reward like one would do in single-agent RL. This method seems to work well in some scenarios [9, 10, 12]. CADIA-Player was one of the first game playing AI to apply this method in non-cooperative games and had taken first place in the third Annual General Game Playing Competition [9]. The work done in [10] has shown the potential of non-cooperative MARL in real-world applications by adopting IQL to help improve efficiency and reduce waste in the channel access problem. Kopazs et al. [12] applied IQL in the predator-and-prey game and found that IQL converges faster in their test, outperforming the other algorithms. Researchers [11] also proposed a more sophisticated method for computing the policy from the Q-values, aiming for convergence towards NEs. However, none of the studies mentioned above had analyzed the behavior of the agents or identify any flaws of IVBL from a game theory point of view. It was formally proven that Q-learning converges with probability 1 in single-agent RL [40]. Although the empirical results appear promising and may serve as an indicator, they do not provide any formal guarantees. We still do not fully understand the convergence of IVBL agents' behavior in non-cooperative games. Formal analysis of the agent's behavior along with a game to verify the analysis may give us more confidence about its convergence rate, or it may reveal weaknesses of the algorithm that was hidden behind these results. Unfortunately, our work found the latter is true, raising an alarm to MARL researchers.

3. MARL preliminaries

In this section, we provide preliminaries including the formal definition of a game and NE.

3.1. Stochastic game

A game is formally defined by a tuple: $G = (N, S, S_0, S_t, \{A_1, \dots, A_n\}, \{R_1, \dots, R_n\}, P)$

- $N = \{1, \dots, n\}$: the non-empty set of agents.
- S : the game's non-empty state space.
- $S_0 \subset S$: the non-empty set of possible initial states.
- $S_t \subset S$: the set of terminal states.
- A_i : the action space for the i^{th} agent. Let $a_i \in A_i$ be an action of agent i , and $a^n = (a_1, \dots, a_n) \in A^n$ denote a joint action in the joint action space. It's worth mentioning that A may contain a 'pass' or 'idle' action where the agent does nothing actively.
- R_i : the reward function for the i^{th} agent. $R_i : S \times A^n \times S \rightarrow \mathbb{R}$.
- P : the state transition probability function. $P : S \times A^n \times S \rightarrow [0, 1]$. It defines the probability of the game transitions from one state to another given the joint action. $P(s, a^n, s') = Pr(s'|s, a^n)$. It must satisfy the axiom of probability:

$$\forall s \in S, \quad a^n \in A^n : \sum_{s' \in S} P(s, a^n, s') = 1 \quad (2)$$

Starting from a randomly chosen $s_0 \in S_0$, at each time step $t \in \mathbb{N}$ with the current state s_t , each agent observes the current state s_t and chooses an action forming a joint action $a^n = (a_1, \dots, a_n)$. The game then moves into the next state s_{t+1} , randomly sampled from the distribution P and a reward r_i is assigned to each agent. If $s_t \in S_t$, the game ends. Otherwise, the game continues in the next time step.

The way a player plays the game defines a policy (strategy) as a function that computes the probability of selecting each action given the current state, $\pi : S \times A^n \rightarrow [0, 1]$. A strategy is said to be pure if it is deterministic ($\pi : S \times A^n \rightarrow \{0, 1\}$). Otherwise, we call it a mixed strategy. A set of strategies, one for each player, is called a profile π^n . Given the profile, we can compute the expected reward for each player in the game. The goal is to find the optimal strategy for an agent, or even the optimal profile, such that expected cumulative rewards are maximized. Cooperative games are special cases where agents share the same reward. The optimal profile is the one that maximizes the common reward. Because they share the same reward function, the problem can be formulated as a joint player selecting a joint action and receives a reward, which makes cooperative games essentially single-agent games from a game theory perspective. In general non-cooperative games, however, it may not be possible to make everyone happy due to the conflicting interests. A more profound definition of optimum need to be discussed in the non-cooperative games section.

Formally, let $R_i^*(\pi^n)$ be the function to compute the expected reward of the i^{th} player given the profile. Let π_i be a policy of the i^{th} player and Π_i be the policy space of the i^{th} player. Let π^{-i} denote $\pi^n - \{\pi_i\}$, the set of strategies of all other players. A profile π^n is a NE if

$$\forall i \in N, \quad \forall \pi_i \in \Pi_i : R_i^*(\{\pi_i\} \cup \pi^{-i}) \leq R_i^*(\pi^n) \quad (3)$$

The idea is that NEs are the stable points in the game, because if an agent can benefit from switching strategy on its own, they would do so.

A game is finite if $|N|$, $|S|$ and all $|A_i|$ are finite, and the game always ends in a finite number of turns (thus, $S_t \neq \emptyset$). John Nash proved that every finite game has at least one NE. To

solve a game in the context of this paper means to find at least one NE. Games that are not finite can be ill-conditioned and have no NE. For example, a game where the player who names the greatest number wins has no best strategy. For the rest of the paper, we only refer to finite games. A normal-form game has only two states, the starting state S_0 and the end state S_1 . The game ends immediately after all agents pick an action. Thus, they can be represented using the reward matrix. In the book on game theory originally published in 1944 [41], Von Neumann and Morgenstern showed that finite games can be translated into normal-form games. If we encapsulate the following states as a subgame, we end up with a reward matrix for the initial state with the rewards being the expected rewards of the subgames. We focus on normal-form games because they are easier to work with for the analysis. Since they only have one state S_0 for agents to take actions, we simply write $\pi(a)$ instead of $\pi(s_0, a)$ to denote the probability of taking action a .

4. Stability of IVBL in non-cooperative games

In this part of the paper, we show the feasibility of IVBL algorithms in non-cooperative games with behavioral analysis. We demonstrate the necessary conditions the game must satisfy in order to apply the IVBL algorithms.

To demonstrate the problem with competitive games, we study two simple normal-form games. Consider the simple one-shot matrix game rock-paper-scissors (RPS) in Table 1(a). Each entry represents the reward for the row player and the column player, respectively. The only NE in the game is that two players play each of the three actions with 1/3 chance (the policy can be represented by a vector $[1/3, 1/3, 1/3]$). If any player deviates from the NE, their expected reward does not increase, and the other player can exploit them by playing the action that counters their most frequent action. This kind of anti-exploitation strategy defined by NEs directly results in a well-known consequence in game theory. At any NE, given the policies of all other players, player i is indifferent from all actions that are chosen with a positive probability [14]. In other words, all actions played by the agent generate the same expected reward. The theorem also states that the actions are split into two sets, the worthy actions that are chosen with a positive probability and the worthless ones that will never be picked. For example, a variation of RPS could incorporate an always-losing action, where the player receives negative reward once executing it, regardless of the opponent's action. Such an action should never be selected by the players. The above fact can be formally stated as the following. Assume profile π^n is a NE. Let $Q_i(a_j, \pi^{-i})$ be the expected reward of action $a_j \in A_i$ for agent i , given the rest of the profile π^{-i} . Let A_i^+ denote the set of worthy actions of agent i , namely $A_i^+ = \{a \mid a \in A_i \wedge \pi_i(a) > 0\}$. We have:

$$\forall a_j, a_k \in A_i^+, Q_i(a_j, \pi^{-i}) = Q_i(a_k, \pi^{-i}) \quad (4)$$

Equation (4) gives us a way to compute NEs in two-player normal-form games. Since $Q_i(a, \pi^{-i})$ (the expected reward generated by action a for player i) can be computed as the dot product of π^{-i} (the opponent's policy) and the row/column in the reward matrix corresponding to that action a , we can setup a set of linear equations so that all actions generate equal expected reward for each agent. Solving the system of linear equations gives us the NE. Table 1(b) defines a more complicated game called biased rock-paper-scissors. Using linear algebra, we can find the only NE in the game is that both players adopt the following

Table 1. The reward matrix.

(a) Rock-Paper-Scissors reward matrix			
	r	p	s
r	0,0	−1,1	1,−1
p	1,−1	0,0	−1,1
s	−1,1	1,−1	0,0
(b) Biased Rock-Paper-Scissors reward matrix			
	r	p	s
r	50,50	25,75	100,0
p	75,25	50,50	45,55
s	0,100	55,45	50,50

Note: It is used to define the reward for each player for each joint action. The row player chooses a row and the column player chooses a column, then the row player gets the first reward and the column player gets the second. Technically speaking, a reward matrix is actually two matrices, one for each player, but they are often written as one for simplicity.

policy stated in Equation (5). This game will serve as an example to show how IVBL can lead to failure in our experiments later.

$$\pi = (Pr(r) = 0.0625, Pr(p) = 0.6250, Pr(s) = 0.3125) \quad (5)$$

Suppose a IVBL algorithm can converge to a NE, then all worthy action's Q-values must converge to the same for an agent. It does not matter how the algorithm computes a probability distribution over the actions at the end, it cannot make any discriminations among those worthy actions based on equal statistics. Such an algorithm will output at best a uniform distribution over the worthy actions at a NE. This contradicts with the assumption that they converge to a NE unless the NE is full of uniformly random policies which is not the case in most of the games. We have shown that pure IVBL approaches does not work through proof by contradiction.

This sets a necessary condition for IVBL algorithms to work. IVBL algorithms can converge to a NE only if the NE consists of uniformly random policies over the set of all worthy actions. Note that this also includes any pure (deterministic) policies. It can be formally stated as follows. Let Π^n be the set of all possible profiles, and Π^{n*} be the subset containing all the NEs.

$$\exists \pi^n \in \Pi^{n*}, \forall \pi_i \in \pi^n, \forall a_j, a_k \in A_i^+, \pi_i(a_j) = \pi_i(a_k) \quad (6)$$

Because in cooperative games, there is no need to use mixed strategies, the NEs are essentially pure. This means that all cooperative games satisfy Equation (6), which explains why IVBL works in cooperative games. Additionally, according to Zermelo's theorem [42], perfect-information two-player finite asynchronous (players take turns) games such as chess or Go also satisfy the condition, because they also have a pure NE.

Then how did the researchers still find the appealing outcome of IVBL in some non-cooperative games [9–12]? The behavior of IVBL can be highly problem-dependent, so we provide only high-level insights here. MARL problems studied in the real world can have large state/action spaces. For example, a 19×19 Go board can have roughly $3^{19 \times 19} = 10^{172}$ states. It may be too difficult for the agents to converge to a global NE because of the limited computational power. Because of the complexity of the environment mechanics

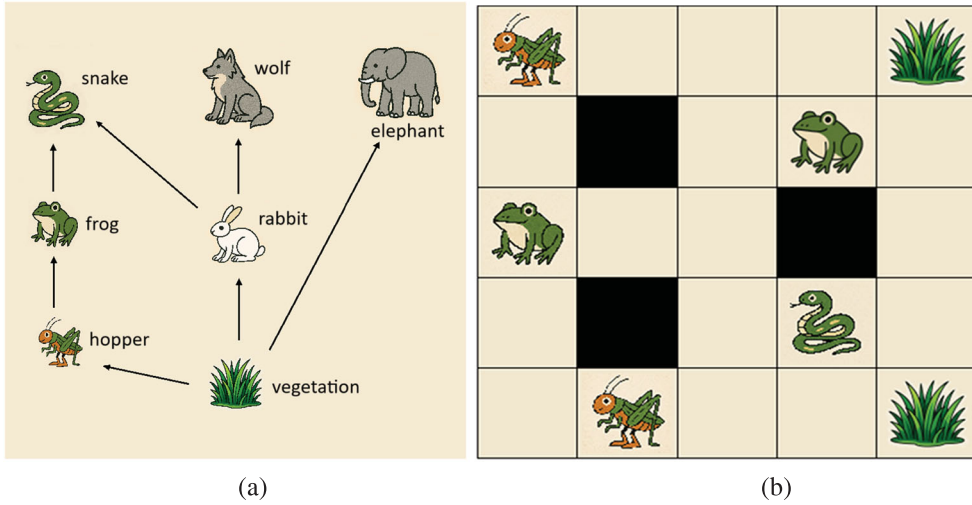


Figure 1. (a), A food web represents a simple ecosystem. The food web is a cluster of multiple food chains showing what-eats-what. The arrows point in the direction of the energy flow. (b), In the Food Chain game, an agent represents an individual animal of some type. They navigate on a playground trying to feed themselves while avoiding enemies.

itself, the competitiveness of the game is not going to play an enormous role before agents can reliably predict the state transitions. Picking the action with the best Q-value still gives reasonably good results, since at least the agents learned to interact with the environment.

5. The food chain game

To verify our analysis, we test the algorithms in a game that can turn on and off the necessary condition defined by Equation (6) described in the previous section. We define the mechanics of the game Food Chain. An agent in the game represents an animal of some type. They can move around the map and feed themselves while trying to avoid enemies. The game simulates an ecosystem to a certain extent, by defining a food web structure. A food web is a cluster of food chains showing what-eats-what in the ecosystem as illustrated in Figure 1. The food web in the game is completely customizable and we can define an arbitrary number of species (agent types) and any number of agents of each type. This game is essentially a generalized version of the classic predator-and-prey problem. Recent research has tested IQL in the predator-and-prey game and found the algorithm to be seemingly performing well [12]. As we stated, these results could be misleading¹.

The game is played on a $w \times h$ grid (in our experiments for example, we used 2×2 and 5×5 scenarios). Each agent is placed in one of the tiles. There are also obstacles placed on the map. These could be randomly generated using a procedural algorithm or pre-defined. The state space S contains all possible configurations of the $w \times h$ game board. A game board configuration includes the current timestep t , as well as the locations of all the agents, vegetation and obstacles. All positions (x, y) must be unique and satisfy $1 \leq x \leq w$, $1 \leq y \leq h$. Any state could be an initial state. A state is terminal if t reaches some preset maximum timestep. The action space $A = \{\text{up, down, left, right, pass}\}$ is the same

for all agents. The state transition function maps the current state and the choices of actions to the next state by executing the agents' actions in a natural way. Agents choose their actions to move along the direction by one cell. Moves resulting in an illegal position will be regarded as pass. If multiple agents try to occupy the same cell, only a randomly selected agent will succeed and the rest will execute the pass action. If an agent is moving away from its current cell, then it is allowed for another agent to enter this cell without conflicts, unless two agents are trying to swap location (this rule prevents agents from passing through each other). The reward function works as follows. After the state transition, if a predator is adjacent to (i.e. within a Manhattan distance of 1) a prey, the prey dies and is removed from the game. Agents receive some negative reward (-1 by default) when they die. Each empty tile also has a chance to grow vegetation each turn. Vegetation is part of the food web and will be destroyed when an agent enter the tile. If the vegetation is a prey to the agent, it will be consumed (this is different from agents eating agents where they only need to be adjacent). Agents get reward (0.1 by default) when they eat. If multiple predators capture a prey at the same time, they must share the reward equally.

We will use this game to demonstrate the weakness of IVBL because the game has several specific advantages that fits this task. The size of the state space grows exponentially with respect to the size of the map and the number of agents. This computational complexity eliminates any tabular method, a brute-force approach that records the value of all possible state-action pairs in a table. In these cases it has been shown that function approximation methods, such as artificial neural networks, should be used [23]. Because of the large state space, it is difficult for us to check whether the agents converged to a global NE. However, the way predators hunt for preys often force agents to address the mixed policy problem, as we will explain in the next section with two example scenarios. We can actually check whether the agents converge to NEs in those subcases. Moreover, the game is able to turn on/off the necessary condition defined by Equation (6) for IVBL to converge so we can tell Equation (6) is really what matters. Thus, the game challenges IVBL particularly against its weak spot and the result can be easily verified.

6. Experimental evaluation

In order to observe how Food Chain can turn on/off the necessary condition defined by Equation (6) for IVBL to converge, we design two scenarios. The key difference between the two is that one satisfies the condition defined by Equation (6) and the other does not. We define a simple food web consisting of frogs and hoppers. Hoppers (represented by h) can consume vegetation (drawn as $*$) and will be eaten by frogs (f). Both scenarios turns off vegetation generation.

6.1. Biased rock-paper-scissors in food chain

This case violates condition defined by Equation (6). The initial state of the scenario consists of a frog trying to hunt down a hopper as shown in Figure 2. Figure 3 shows all possible next states. We can see that in some cases the hopper can escape. If they both pass, they will face the exact same situation again. We set the max timestep to 1, so the scenario becomes a normal-form game. To break the symmetry so that condition defined by Equation (6) is violated, we add vegetation in one cell, making the scenario essentially biased RPS. We also

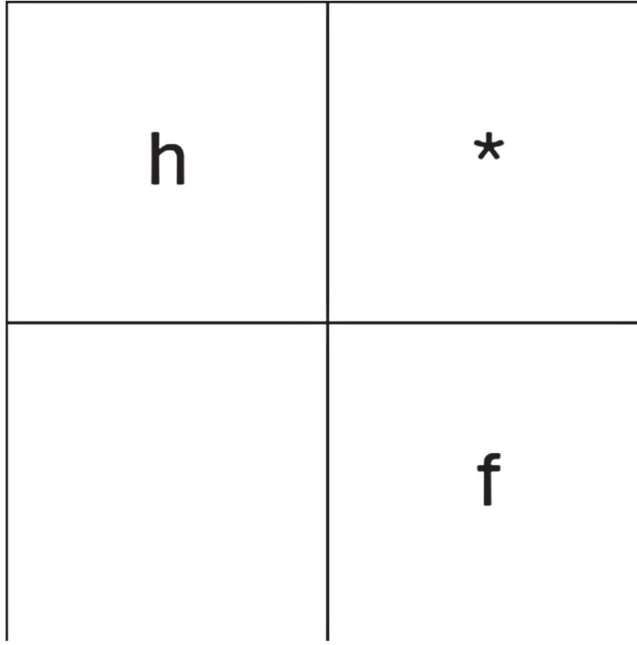


Figure 2. The initial state of Biased RPS in Food Chain. The frog (f) is trying to capture the hopper (h). The only NE in the game contains non-uniformly random mixed policies.

Table 2. Biased RPS in Food Chain reward matrix.

	left	up	pass
right	0.5,0	−0.75,0.5	−0.5,0.5
down	−1,0.5	0,0	−1,0.5
pass	−1,0.5	−1,0.5	0,0

Note: Hopper is the row player.

raise the feeding reward for both agents to 0.5 to highlight the asymmetry. The reward matrix is shown in Table 2. We can verify that there is a unique NE in the game.

$$\pi_h = \left(Pr(r) = \frac{1}{3}, Pr(d) = \frac{1}{3}, Pr(p) = \frac{1}{3} \right) \quad (7)$$

$$\pi_f = \left(Pr(l) = \frac{1}{13}, Pr(u) = \frac{6}{13}, Pr(p) = \frac{6}{13} \right) \quad (8)$$

The expected reward for the hopper and the frog should be $-7/13 \approx -0.538$ and $1/3 \approx 0.333$, respectively.

In the Food Chain game, because all agents have the same mobility, a predator can only capture a prey by trapping them in a corner. Thus, any predator, unless working in a team, has to face this subcase at some point during a general Food Chain game played on an open ground. The scenario represents a subgame which shows up frequently in a full game.

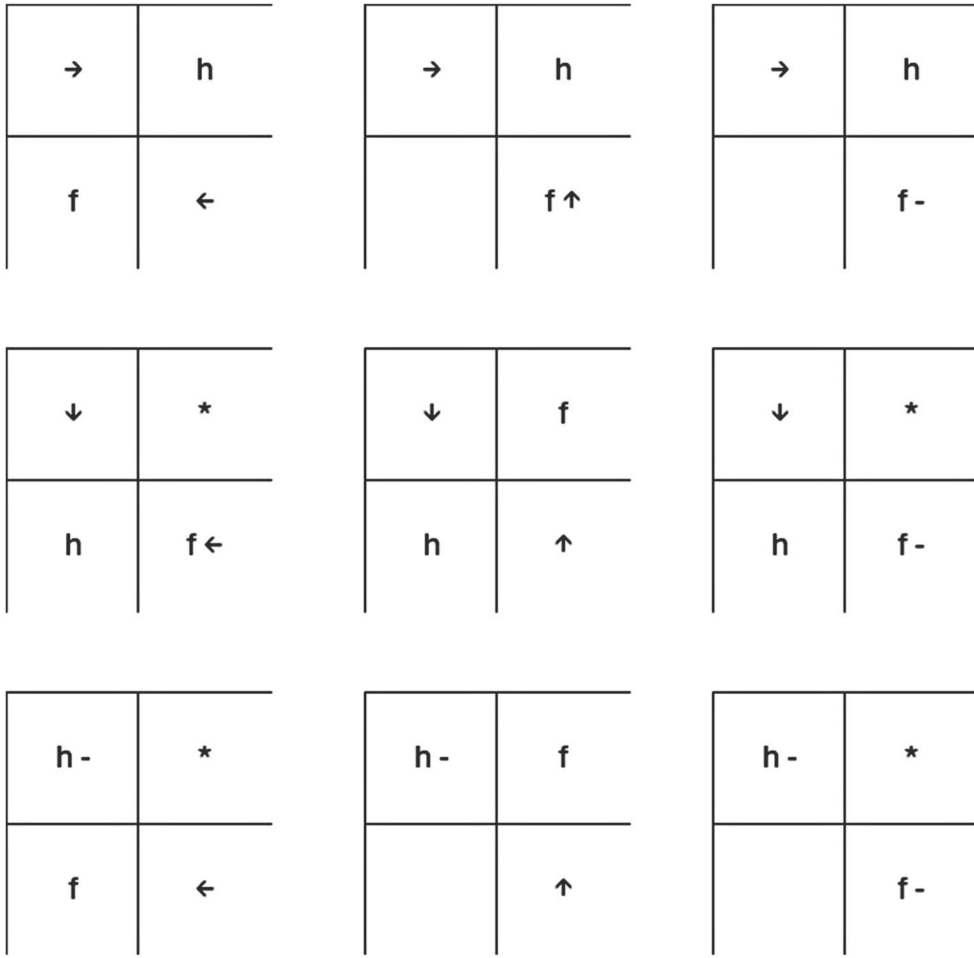


Figure 3. Possible Next States in Biased RPS in Food Chain (omitting illegal moves). The arrows sit in the agent's previous position and point in the direction of their move. Note that if both agents try to move into the vegetation tile, the hopper only has a 50% chance to get the vegetation.

6.2. Tree maze chase in food chain

This scenario essentially turns off the mixed policy problem. We set up the initial state as shown in Figure 4 where a frog is chasing a hopper in a maze. This map resembles a tree structure (no cycles) which creates a pure NE, because there is always an optimal action. The hopper should go for the tunnel with more vegetation and the frog should simply chase the hopper. The necessary condition defined by Equation (6) is satisfied. To make sure that at most two vegetations can be picked up by the hopper, we set the max timestep to 12. The optimal rewards for the hopper and the frog are -0.8 and 1 , respectively.

6.3. The IVBL algorithm

The pseudocode in Algorithm 1 presents the Independent Q-Learning algorithms tested in our game. The notation refers to the definition in Section 3.1. The implementation of the

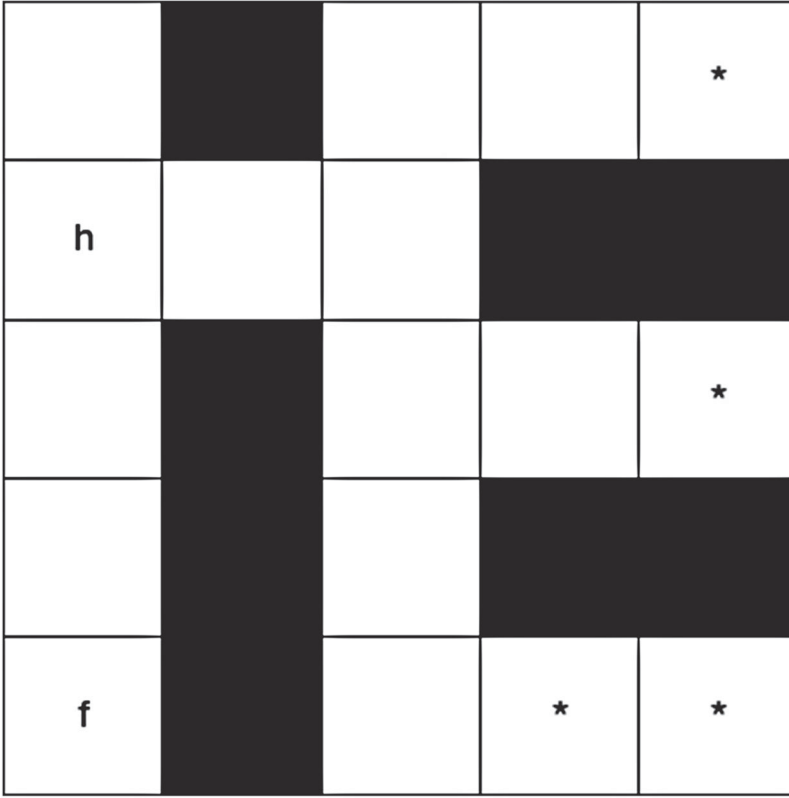


Figure 4. The initial state of Tree Maze Chase in Food Chain. The frog (f) is chasing the hopper (h) in the maze. There is an optimal pure (deterministic) policy for both players.

algorithm is sourced from the open-source MARL library BenchMARL [43]. Their implementation used an artificial neural network to approximate the Q-values. ε -greedy exploration scheme was used with decaying ε . More details of the experiment setup can be found in the appendix.

6.4. Results and discussion

We tested the IVBL algorithm IQL as well as an independent policy-based algorithm IPPO for comparison in both scenarios. The experiments were run using the open-source MARL library BenchMARL [43]. All experiments ran for 800 iterations. For each iteration, the algorithm collects 640 timesteps as training samples and train the neural network before 50 evaluation games were run. The plots show the average reward for each agent during the evaluation phase. Each experiment was run 20 times to mitigate the effect of randomness and the graphs show the boots trapped 95% confidence interval. More details of the parameters for the experiments can be found in Table A1 in the appendix.

Figure 5 demonstrates the performance of the algorithms in the biased RPS scenario where the necessary condition defined by Equation (6) for IVBL to converge is not met. The results show that the IVBL algorithm IQL failed to converge to a defined solution range and

Algorithm 1 Independent Q-Learning

```

 $\forall i \in N, s \in S, a \in A_i, Q_i(s, a) \leftarrow 0$ 
for episode  $k = 0$  to  $\infty$  do
  reset game to an initial state  $s_0 \in S_0$ 
  for timestep  $t = 0$  to  $max\_timestep$  do
    observe current state  $s_t$ 
    with probability  $\varepsilon$ : uniformly randomly choose an action  $a_{i,t} \in A_i$ 
    otherwise: choose  $a_{i,t} = \arg \max_{a_i} Q_i(s_t, a_i)$ 
    (all other agents  $j \neq i$  also pick their actions)
    receive reward  $r_{i,t}$  and observe next state  $s_{t+1}$ 
     $Q_i(s_t, a_{i,t}) \leftarrow \alpha[r_{i,t} + \gamma \max_{a_i} Q_i(s_{t+1}, a_i)] + (1 - \alpha)Q_i(s_t, a_{i,t})$ 
    if  $s_{t+1} \in S_t$  then
      break
    end if
  end for
end for

```

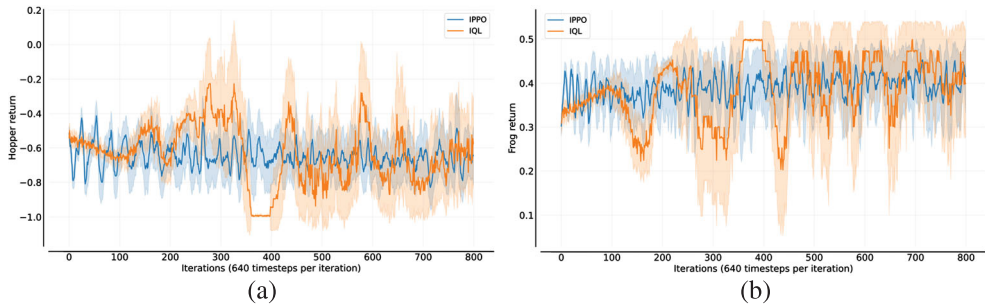


Figure 5. Biased RPS. IVBL (IQL) shows significantly more instability and does not converge to the NE. (a) Biased RPS Hopper Rewards and (b) Biased RPS Frog Rewards.

were struggling to stabilize itself. The hopper’s reward fluctuates between -1 and -0.2 , where the optimal value is -0.538 . These value can be as far as 85% away from the desired amount. The frog reward oscillates in the range $[0.2, 0.5]$, which is in a $\pm 70\%$ range from the actual NE (0.333). For comparison, the independent policy-based learning algorithm IPPO achieves a reward that stays more in the optimal range and is far more stable which indicates the agents are converging towards the NE.

In contrast, IVBL does significantly better in the tree maze scenario, as shown in Figure 6. This time, the game scenario satisfies the condition defined by Equation (6). Although still unstable, IQL can roughly reach the optimal reward at the end. The optimal rewards for the hopper and the frog are -0.8 and 0.1 , respectively. Even in this case, the policy-based method still appears to be more reliable than IVBL.

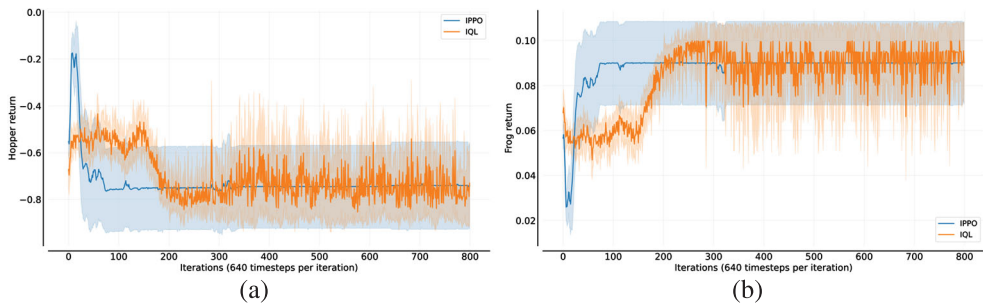


Figure 6. Tree Maze Chase. Both algorithms quickly learned to play the game and tend to stay in the optimal reward range. Note that increasing frog reward reduces the hopper reward as the game is non-cooperative. (a) Tree Maze Chase Hopper Rewards and (b) Tree Maze Chase Frog Rewards.

7. Conclusions and future work

In this work, we have analyzed the stability of independent value-based learning (IBVL) algorithms in non-cooperative games and tested their performance in specific game scenarios. By means of proof by contradiction and the design of experiments based on the proposed Food Chain game, we found that the studied algorithms only converge to Nash equilibrium (NE) in non-cooperative settings when every agent's policy evenly chooses the next action among their worthy actions (actions that are played with a positive probability). This explains why promising results are achieved when applying IVBL to games with pure NE, including cooperative games.

This work also proposes the Food Chain game which has certain traits –such as ease of navigation, focus on interactions, and extendability, among others– that are instrumental for the analysis of the stability in non-cooperative settings. The test results show that IVBL algorithms perform poorly in scenarios where the identified condition is violated. Our work is limited to small two-player scenarios in the Food Chain game. There are still interesting characteristics of the game that will be explored in the future. The game is extendable in many dimensions, the number of agents of each type, the number of species, the relationship between the species, and the size of the map. This opens up room for future research directions. The game can also be viewed as a simplified model for ecosystems to study problems such as wildlife conservation. We can explore how the complexity and the structure of the food web affect the population and behaviors of agents. As the game is simple enough to be human playable, it may also be interesting to see how IVBL agents interact with human players who are not typically Nash players.

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Note

1. For visual illustration, the individual animal avatars were generated by ChatGPT GPT-4-turbo model which includes image generation via DALL·E 3, provided by OpenAI.

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Appendix. Experiments parameters

Table A1. The parameters used in the experiment.

Game parameters		
	Biased RPS	Tree Maze Chase
Size of the map	2×2	5×5
Max timestep	1	12
Feeding reward	0.5	0.1
Death reward		-1
Vegetation spawn chance		0
Number of agents		2
Agent types		2
Experiment parameters		
Repetitions of each experiment		20
Number of iterations		800
Evaluation games after each iteration		50
Training hyper parameters		
	IPPO	IQL
On-policy minibatch optimization iterations ^a	10	
Off-policy optimizer steps ^b		100
Off-policy memory buffer size		20,000
Initial/end ϵ parameter for ϵ -greedy action selection		0.8/0.1
Learning rate α		5×10^{-5}
Discount factor γ		0.99
Frames collected for each iteration		640
Minibatch size		32
The ϵ parameter of the adam optimizer		1×10^{-6}

Note: ^aThe number of times collected frames will be split into minibatches and trained. ^bThe number of times a minibatch will be sampled from the memory buffer and trained over.